

Greece¹¹**Hellenic Competition Commission****I. Introduction**

Effective analysis of email communication plays a critical role in detecting cartel behavior, particularly in competition law enforcement, where inter-company correspondence can provide key evidence. To strengthen its investigative capabilities, the Hellenic Competition Commission (HCC) has adopted specialized commercial software for the analysis of seized data in its workflows (e.g., Tovek Tools). Complementing the existing technical infrastructure, the Forensic Investigations Unit of HCC has recently initiated the internal development of a system designed to automatically detect direct email exchanges between companies implicated in a dawn raid.

This initiative aims to support case handlers in navigating large volumes of data during the early stages of an investigation under Article 101 of the Treaty on the Functioning of the European Union (TFEU), by efficiently identifying potentially collusive communication among competitors. The system is currently in a pilot phase, undergoing evaluation in two real-world cases following the completion of its initial development.

II. Related Work

The increasing volume of digital evidence in antitrust and competition law cases has emphasized the need for effective email analysis and communication forensics. Prior research has explored the application of both supervised and unsupervised machine learning models to classify email content, graph-based network analysis to identify suspicious communication patterns, and Technology-Assisted Review (TAR) tools for document review in e-discovery processes.

A key challenge in this area is the extraction of structured data from unstructured sources such as emails. Studies have highlighted challenges in entity resolution (i.e., linking individuals to specific organizations), domain attribution (i.e., associating email domains with known organizations), and semantic interpretation (i.e.,

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understanding contextual meaning in communication). However, relatively few research efforts focus on the detection of cartel-related communications, despite the significant legal and economic consequences. This work builds on existing well-established approaches by introducing a domain-specific, modular, and robust system that integrates structured extraction, investigative logic, and visual analytics to support competition law enforcement in cartel investigations.

III. Methodological Framework

A. System Design

The system consists of five core components, as illustrated in Figure I:

- **Data Extraction:** Structured information (metadata) is extracted from email evidence files (MSG format).
- **Data Ingestion:** The extracted raw data is loaded into a relational database for structured storage, access, and querying.
- **Business Intelligence Processing:** SQL-based business logic is applied to associate emails with investigated companies and identify inter-company communications.
- **Data Export:** Relevant information is exported in analyst-friendly formats (e.g., Excel) to support further case analysis.
- **Data Visualization:** The communication pattern is visualized as a network graph to provide an instant understanding of interactions between companies.

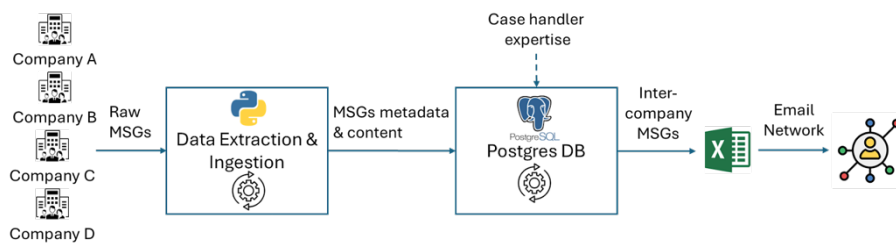


Figure I: General system overview

B. Information Flow

The system's input consists of the electronic communications (emails) collected during a dawn raid. The **Data Extraction** module, implemented in Python, parses MSG files to extract both content and key metadata, including sender, recipients, subject, date, and number of attachments. Additionally, it isolates the

sender's email address and domain into distinct fields to enable granular analysis of email affiliation. The primary objective at this stage is to transform the raw, unstructured data of emails into a structured format suitable for subsequent analysis.

In the **Data Ingestion** phase, the extracted information is stored into a PostgreSQL database using Python. PostgreSQL was selected for its robust support for structured storage, complex querying, and efficient data retrieval capabilities. At this stage, no filtering is applied; instead, the system creates a structure to preserve all the parsed information to ensure flexible investigation.

The **Business Intelligence** component is responsible for applying domain-specific logic written in SQL and focuses on deriving meaningful insights for case handlers. A central step in the process is the association of each email with one or more companies under investigation. Ultimately, this association enables the identification and mapping of inter-company communication patterns, which is essential for detecting potential cartel formation.

Nevertheless, not all email addresses can be reliably linked to a known corporate domain. This is particularly challenging when communications originate from generic email providers (e.g., *@gmail.com). To mitigate this inherent ambiguity and improve accuracy, the system allows the optional integration of expert knowledge. In this way, if a case handler confirms that a previously unknown email address belongs to a specific company, this input is recorded and incorporated back into the system, thus reducing uncertainty and refining classification.

The output, currently exported as an Excel file, includes selected emails along with their metadata, company associations, and a certainty score. This score indicates the likelihood that the communication is inter-company: a score of 1 denotes a definitive match, while 0 represents a possible, yet unverified connection. This structured output provides a clear and interpretable view of communication relevance and can serve as the basis for deeper investigative analysis.

Finally, the **Data Visualization** module leverages Python libraries to generate a graph-based representation of the communication network, offering a high-level overview of interactions between companies. This visual summary facilitates the rapid identification of potentially collusive communication patterns (see Figure II).

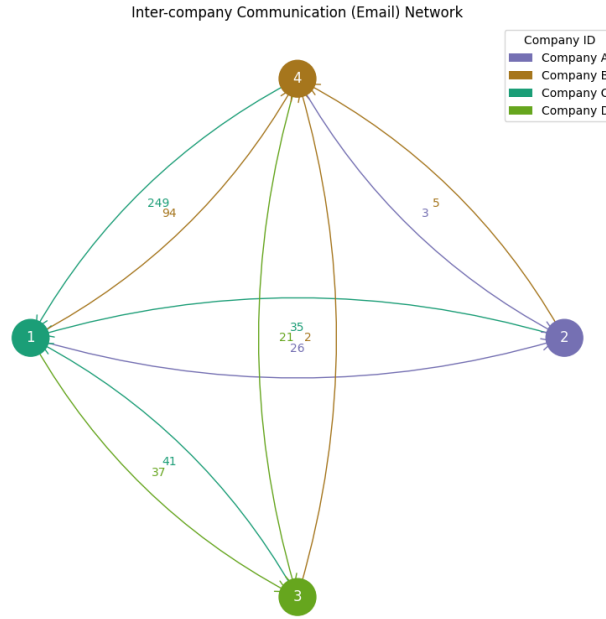


Figure II: Indicative graphical representation of email communications

C. Limitations

Despite its promising functionality, the system faces several technical limitations. One of the most significant challenges is automatic and reliable entity resolution, particularly in accurately linking personal or generic email addresses to specific companies. This is also related to the risk of false positives, where emails flagged as inter-company communications may not, in fact, represent such interactions. Another constraint is the current lack of semantic understanding, meaning that while structural patterns are detected, the actual intent behind messages is not yet interpreted to distinguish collusive behavior from legitimate exchanges. Furthermore, the scalability of the system is still under evaluation; performance with very large datasets may require further optimization or architectural upgrades.

IV. Conclusion and Future Work

This system represents an initiative by HCC to support the identification of email exchanges between investigated companies acting as competitors. By transforming unstructured email data into a structured and analyzable format, the system enhances the efficiency of early-stage investigations. Its current implementation relies on deterministic, rule-based logic and keyword filtering to locate potentially relevant communications.

While effective for initial triage, keyword-based searching can be limited in scope, often missing semantically related content or flagging irrelevant messages. To address these limitations, future development will focus on expanding the system's analytical depth through semantic search capabilities since the identification of communications based on conceptual meaning may be useful in detecting subtle or coded language often used in cartel-related correspondence. This enhancement will enable more detailed classification of emails as relevant or irrelevant for the cases, further assisting case handlers in their investigative efforts.